Practice

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

- Practice Assessment 2 (FSM209P)
- Complete step-by-step worked models
- Official Mark Scheme Allocations [M1 / A1]
- Exam Board: Open Awards (Ofqual Regulated)

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LEVEL 2 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION A - QUESTION AND ANSWER PAPER
NON-CALCULATOR – 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM209P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section A** at the end. Once you have handed in this question and answer paper, you will not be able to check this again.
7. If you use extra paper, write your name, learner number and the question number you are answering on it and securely attach it to this question and answer paper.

Learner name:	
Learner number:	
Centre number:	
Signature:	
Today's date:	

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Section A

Question 1

What is 4.2 as a fraction of 33.6? (1 mark)

Show your calculations and/or workings out here:

Step 1: Write as fraction: 4.2 / 33.6

Step 2: Multiply by 10 to clear decimals = 42 / 336

Step 3: Simplify fraction by dividing by 42:

42 ÷ 42 = 1, 336 ÷ 42 = 8 --> 1 / 8 [1 mark]

Write your answer in this box:

1 / 8 ✓

Question 2

Calculate the pressure exerted by a force of 8000 newtons on an area of 25m².
(2 marks)

Pressure = Force ÷ Area

Show your calculations and/or workings out here:

Step 1: Pressure = Force ÷ Area = 8000 N ÷ 25 m² [M1]

Step 2: Bus stop division: 8000 ÷ 25 = 320 N/m² [A1]

```
   3 2 0
25 | 8 0 5 0 0
```

- 80 ÷ 25 = 3 remainder 5
- 50 ÷ 25 = 2, 0 ÷ 25 = 0 --> 320

Write your answer in this box:

320 ✓ newtons/m²

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Question 3

Put the following fractions in order of size: (1 mark)

$\frac{7}{3}$ $\frac{5}{9}$ $\frac{7}{10}$ $\frac{3}{7}$ $\frac{4}{5}$

Write your answer in this box:

• Decimal benchmarks: 3/7 ≈ 0.429, 5/9 ≈ 0.556, 7/10 = 0.700, 4/5 = 0.800, 7/3 = 2.333

3/7, 5/9, 7/10, 4/5, 7/3 ✓ [1 mark]

Question 4

There are two red pens, eight purple pens and six yellow pens in a box.

One pen is chosen at random from the box.

What is the probability that the chosen pen is red or purple? Give your answer as a percentage.

(2 marks)

Show your calculations and/or workings out here:

Step 1: Total pens = 2 (red) + 8 (purple) + 6 (yellow) = 16 pens

Step 2: Number of red or purple pens = 2 + 8 = 10 pens

Step 3: P(Red or Purple) = 10 / 16 = 5 / 8 = 0.625 [M1]

Step 4: Percentage = 0.625 × 100 = 62.5% [A1]

Write your answer in this box:

62.5% ✓

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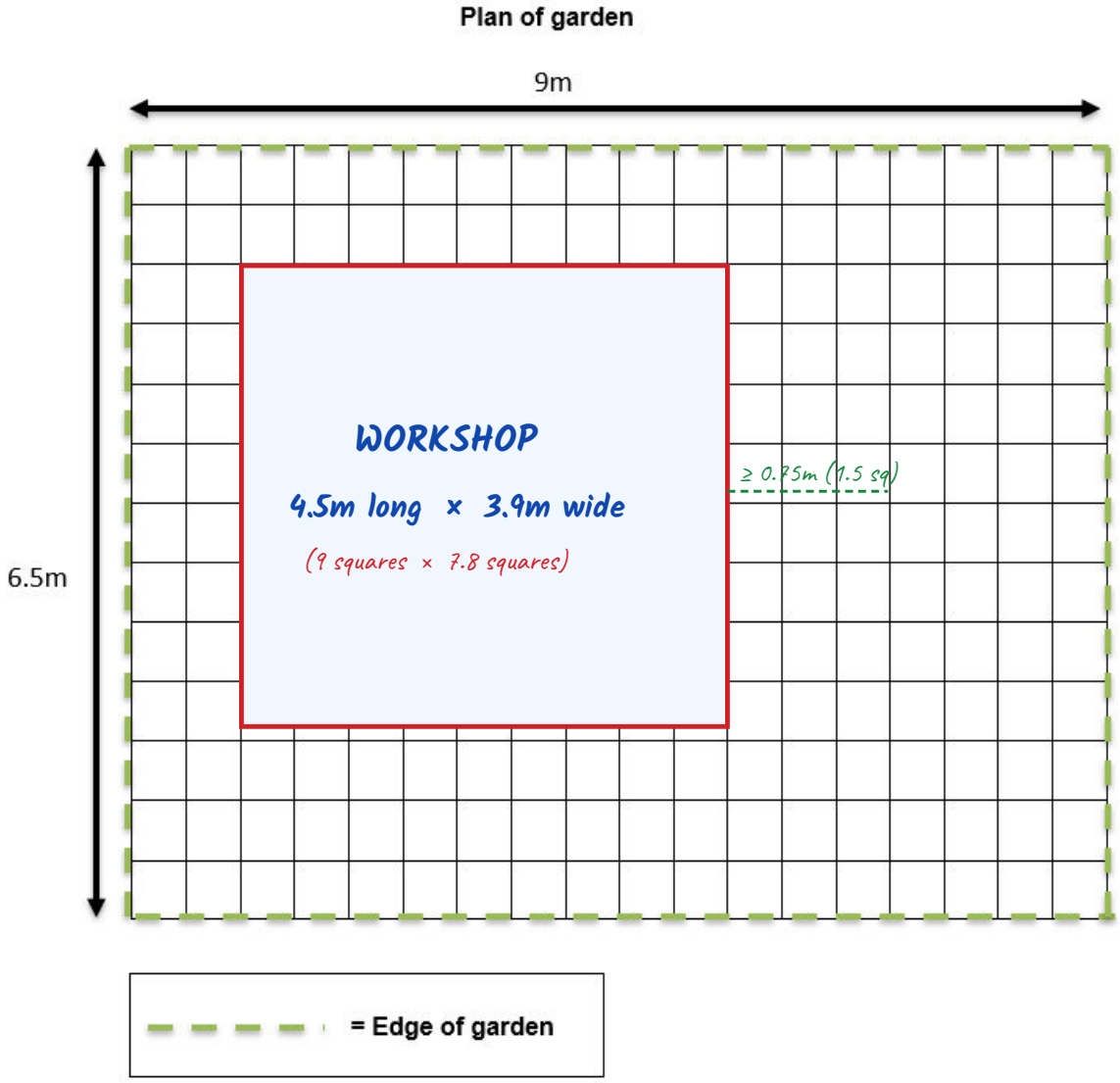
Question 5

John wants to build a workshop in his garden. The workshop will be rectangular, 4.5m long by 3.9m wide. The workshop must be **at least** 0.75m from the edge of the garden.
Add a scale drawing of the workshop to the plan of the garden below:

(3 marks)



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Show your calculations and/or workings out here:

Step 1: Calculate grid scale from garden dimensions:

Garden length = 9m spanning 18 grid squares

Scale = 9m ÷ 18 = 0.5m per square (or 1m = 2 squares) **[M1]**

Step 2: Convert workshop dimensions into grid square units:

- **Length = 4.5m ÷ 0.5m = 9 squares** **[M1]**
- **Width = 3.9m ÷ 0.5m = 7.8 squares**

Step 3: Calculate minimum border clearance:

- **Minimum distance = 0.75m ÷ 0.5m = 1.5 squares from garden edge**

Step 4: Scale drawing summary:

Workshop rectangle is drawn 9 squares by 7.8 squares on Page 8 plan, positioned with more than 1.5 squares clearance from all garden edges. **[A1]**

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Question 6

Rana wants to buy a new mobile phone. The cheapest price she has found in the UK is £260, but Rana's uncle found the same phone for $310 in the USA.

Rana thinks the mobile phone bought in the UK is cheaper.

Is she correct? Give a reason for your answer. (3 marks)

$1 = £0.80

Show your calculations and/or workings out here:

Step 1: Convert USA phone price ($310) into GBP (£):

Exchange rate: $1 = £0.80

USA price in GBP = $310 × 0.80 = £248.00 [M1]

Step 2: Alternative method (convert UK £260 to USD):

UK price in USD = £260 ÷ 0.80 = $325.00 ($325 > $310)

Step 3: Compare costs and calculate price difference:

UK price (£260) vs USA price in GBP (£248):

Difference = £260 - £248 = £12.00 cheaper in the USA [M1]

Write your answer and reason in this box:

No, Rana is not correct. ✓ [A1]

The phone in the USA costs £248 ($310 × 0.80 = £248).

The UK price is £260, so the USA phone is £12 cheaper.

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Question 7

Alison receives an itemised phone bill for her work phone.

Call	Call duration	Call charges (£)
Call 1	00:01:31	0.195
Call 2	00:00:33	0.074
Call 3	00:01:22	0.126
Call 4	00:00:22	0.030
Call 5	00:01:46	0.210
Call 6	00:47:19	0.000
Call 7	00:32:58	0.000
Call 8	00:05:59	0.814
Call 9	00:01:19	0.123
Call 10	00:00:00	0.000
Call 11	00:00:02	0.000
Call 12	00:04.36	0.000
Subtotal		**1.862**

There is an error in the **subtotal** for the call charges.

How much extra has Alison been charged? (3 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total call charges by vertical column addition:

$$\begin{array}{r} 0.195 \\ 0.074 \\ 0.126 \\ 0.030 \\ 0.210 \\ 0.814 \\ +\,0.123 \\ \hline 1.572 \\ {\scriptstyle 2} \end{array}$$

Step 2: Subtract call charges from total bill:

Total bill = £1.862

Extra charge = £1.862 − £1.572 [M1 A1]

= £0.29 (or 29p) [A1]

Write your answers in this box:

£0.29 (or 29p) ✓

[End of Section A]

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LEVEL 2 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER CALCULATOR – 1 HOUR 30 MINUTES PRACTICE ASSESSMENT 1 (FSM209P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED
You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

Learner name:	
Learner number:	
Centre number:	
Signature:	
Today's date:	

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Section B

Question 8

463 students take an exam and 19 fail. What percentage of students fail the exam?

(2 marks)

Show your calculations and/or workings out here:

Step 1: Write proportion: (19 ÷ 463) × 100 [M1]

Step 2: 19 ÷ 463 = 0.0410367...

= 4.10367...% --> 4.1% (to 1 d.p.) [A1]

Write your answer in this box:

4.1% ✓

Question 9

Put the following decimals in order of size: (1 mark)

2.122 2.962 22.36 22.8 2.112

Write your answer in this box:

• Compare place values: 2.112 < 2.122 < 2.962 < 22.36 < 22.80

2.112, 2.122, 2.962, 22.36, 22.8 ✓ [1 mark]

Question 10

Marta owns a property that she rents out to tenants. The yearly rental income she receives is less than the median average yearly rental income of other similar rental properties in the area.
Marta currently receives a rental income of £795 per calendar month.

Properties in local area	Yearly rental income (£)
Property 1	8250
Property 2	12500
Property 3	11100
Property 4	10500
Property 5	9200
Property 6	10200
Property 7	9900
Property 8	11700

By what percentage is Marta's yearly rental income **below** the median average yearly rental income of the other properties?

(5 marks)

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Show your calculations and/or workings out here:

Step 1: Order the 10 local yearly rental values from smallest to largest:

£8,400, £9,000, £9,600, £9,900, £10,200, £10,500, £11,100, £11,400, £12,000, £12,600

^ *Middle two values (5th & 6th)*

Step 2: Median = (£10,200 + £10,500) ÷ 2 = £10,350 [M1 A1]

Step 3: Marta's yearly rent = 12 × £795 = £9,540 [A1]

Step 4: Difference below median = £10,350 − £9,540 = £810

Step 5: Percentage below median = (Difference ÷ Median) × 100

= (£810 ÷ £10,350) × 100 [M1]

= 0.07826... × 100 = 7.826...% --> 7.8% (or 8%) [A1]

Write your answer in this box:

7.8% *(or 7.83%)* ✓

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Question 11

Marta puts £5000 of the income earned from renting out her property into a savings account paying compound interest at a rate of 2.4% per annum.
How much will Marta have in the account after two years?

(3 marks)

Show your calculations and/or workings out here:

Step 1: Calculate Year 1 interest and balance:

Year 1 Interest = £5,000 × 0.024 = £120.00 [M1]

Balance at end of Year 1 = £5,000 + £120 = £5,120.00

Step 2: Calculate Year 2 compound interest on new balance:

Year 2 Interest = £5,120 × 0.024 = £122.88 [M1]

Balance at end of Year 2 = £5,120 + £122.88 = £5,242.88

Step 3: Multiplier check: £5,000 × $(1.024)^2$ = £5,242.88 [A1]

Write your answer in this box:

£5,242.88 ✓

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Question 12

Marta wants to purchase another property to rent out. She has £61500 cash for the deposit and the house she wants to buy is valued at £192300.

Marta can borrow the outstanding amount on a mortgage providing her loan-to-value (LTV) is below 75%. LTV is calculated using the following formula:

Loan-to-value (LTV) = Amount borrowed ÷ house value × 100

Is Marta able to purchase the property? Give a reason for your answer.

(4 marks)

Show your calculations and/or workings out here:

Step 1: Amount borrowed = £192,300 - £61,500 = £130,800 [M1]

Step 2: LTV formula = (£130,800 ÷ £192,300) × 100 [M1]

Step 3: LTV = 0.680187... × 100 = 68.02% (or 68%) [A1]

Write your answer and reason in this box:

Yes, Marta can borrow the outstanding amount. [A1]

Her LTV is 68.02% (or 68%), which is comfortably below the maximum allowable mortgage limit of 75% (by approx 7%).

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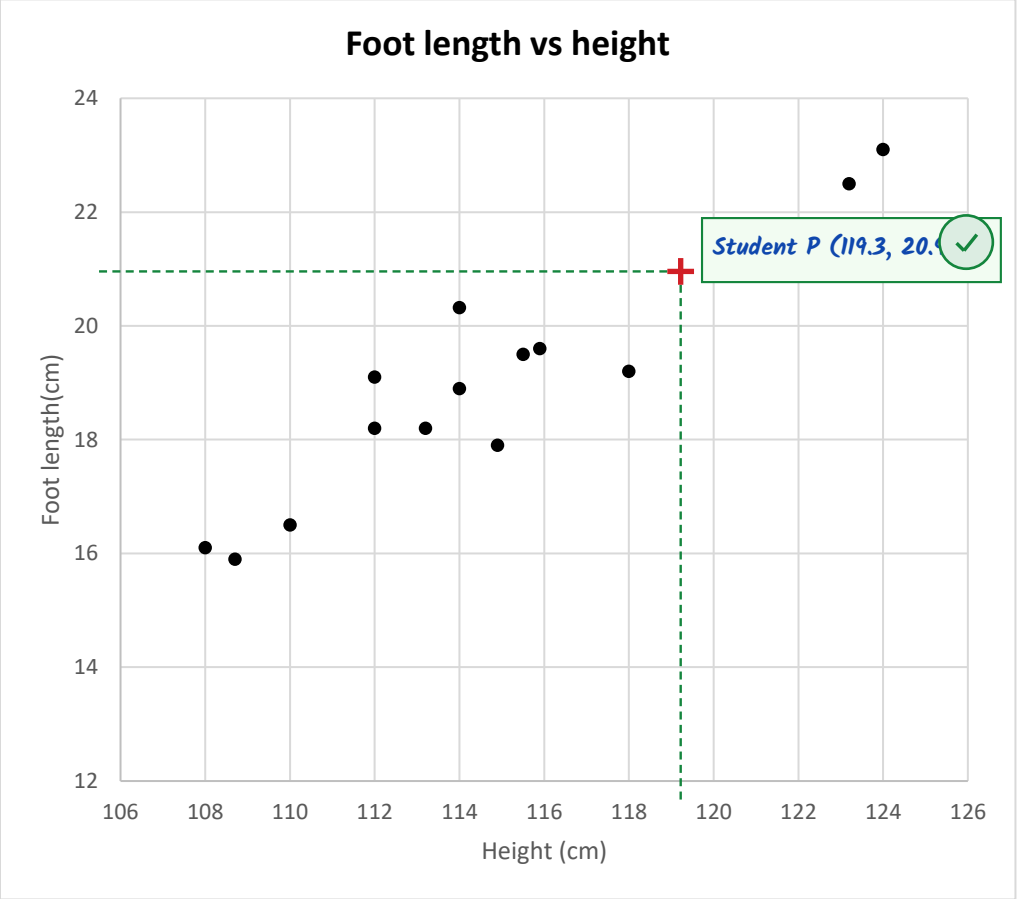
Question 13

The scatter graph below shows the height and foot length of 15 males:



The table below shows information about Student *P* :

Foot Length (cm)	Height (cm)
20.9	119.3

Add student *P*'s data to the graph. (1 mark)

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Question 14

Calculate 3.315 × 7.991 (1 mark)

Show your calculations and/or workings out here:

Calculate product using calculator:

3.315 × 7.991 = 26.490165 *[1 mark]*

(Accepted rounded to 2 d.p.: 26.49)

Write your answer in this box:

26.490165 ✓

For Markers Use Only

Question 15

What is $\frac{7}{4}$ expressed as a decimal? (1 mark)

Show your calculations and/or workings out here:

Convert improper fraction to decimal by dividing numerator by denominator:

7 / 4 = 7 ÷ 4 = 1.75 (or 1 3/4 = 1 + 0.75 = 1.75) [1 mark]

Write your answer in this box:

1.75

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Question 16

A gym asks its members to complete a customer satisfaction survey.

One of the questions asks how many times they used the gym last month.

The frequency has been partially completed. The remaining answers are shown below:

16	**12**	**16**	**21**	**12**	
29	**23**		**18**		**10**
	24	**20**	**19**	**11**	
		16			**20**
21	**17**		**20**	**21**	

Number of times the gym was used last month	**Frequency**	*Midpoint (x)*
0-4	6	2
5-9	5	7
10-14	4	12
15-19	6	17
20-24	8	22
25-29	1	27

Calculate an **estimate** of the mean number of gym visits.

(6 marks)

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Show your calculations and/or workings out here:

Step 1: Calculate midpoints (x) and f × x products for each group:

- Group 0 – 4: Midpoint = 2, f × x = 6 × 2 = 12
- Group 5 – 9: Midpoint = 7, f × x = 5 × 7 = 35
- Group 10 – 14: Midpoint = 12, f × x = 4 × 12 = 48
- Group 15 – 19: Midpoint = 17, f × x = 6 × 17 = 102
- Group 20 – 24: Midpoint = 22, f × x = 8 × 22 = 176
- Group 25 – 29: Midpoint = 27, f × x = 1 × 27 = 27

[M1 A1]

Step 2: Total visits (Sum of fx):

Total visits = 12 + 35 + 48 + 102 + 176 + 27 = 400 visits [M1 A1]

Step 3: Estimated Mean = Total visits ÷ Total members:

= 400 ÷ 30 = 13.333... visits --> 13.3 visits (or 13) [M1 A1]

Write your answer in this box:

Estimate of the mean number of gym visits = 13.3 visits

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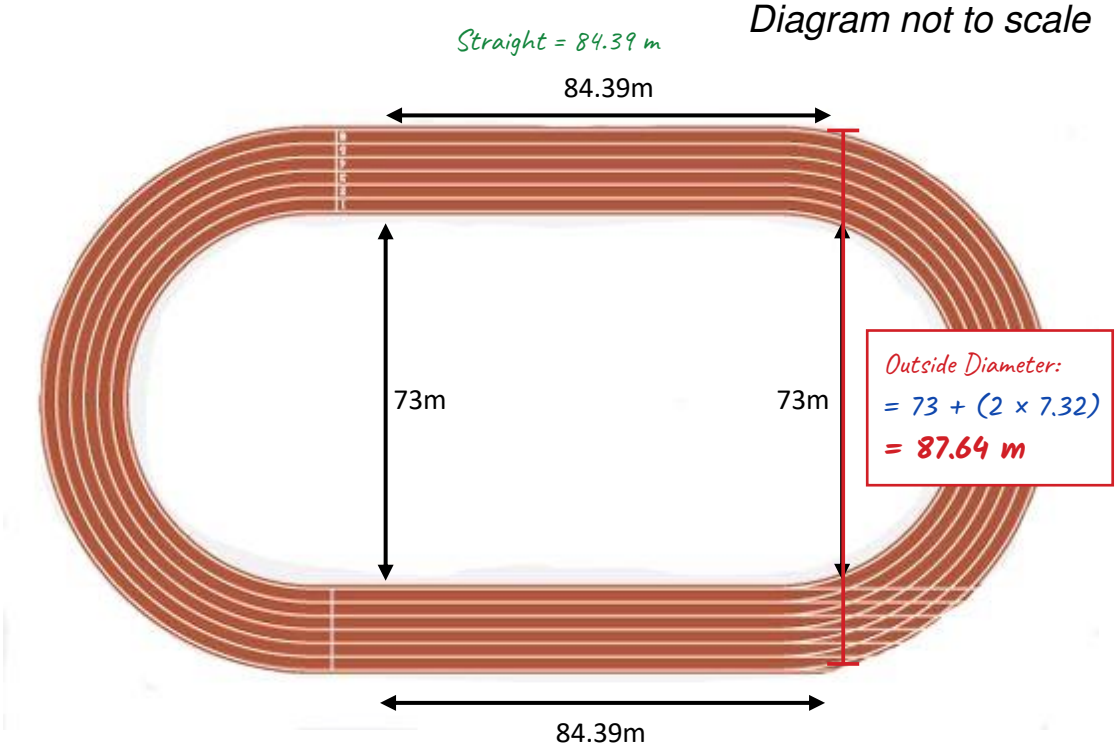
Question 17

The gym has an outdoor running track as shown in the diagram below:

Diagram not to scale



The running track has six lanes and each lane is 1.22m wide.

Holly runs along the **outside edge** of the running track. She wants to run a minimum of 1200m.

π = 3.142

Will Holly run 1200m if she completes three laps of the track? Explain your answer. (6 marks)

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Show your calculations and/or workings out here:

Step 1: Calculate total outside diameter of the track:

Additional width for 6 lanes = 6 × 1.22m = 7.32m per side

Outside diameter = 73m + (2 × 7.32m) = 87.64m [M1]

Step 2: Calculate perimeter of outside track:

- **Two straight sections = 2 × 84.39m = 168.78m**
- **Curved ends (full circle) = π × d = 3.142 × 87.64m = 275.365m** [M1]

Total perimeter = 168.78m + 275.365m = 444.145m [A1]

Step 3: Calculate distance completed in 3 laps:

Distance for 3 laps = 3 × 444.145m = 1,332.4m (> 1200m) [M1]

(Laps needed for 1200m: 1200m ÷ 444.145m = 2.70 laps) [A1]

Write your answer and explanation in this box:

Yes, Holly will complete a minimum of 1200m. [A1]

3 laps around the outside edge is 1,332.4m, which is greater than 1200m (she only needs 2.7 laps to reach 1200m).

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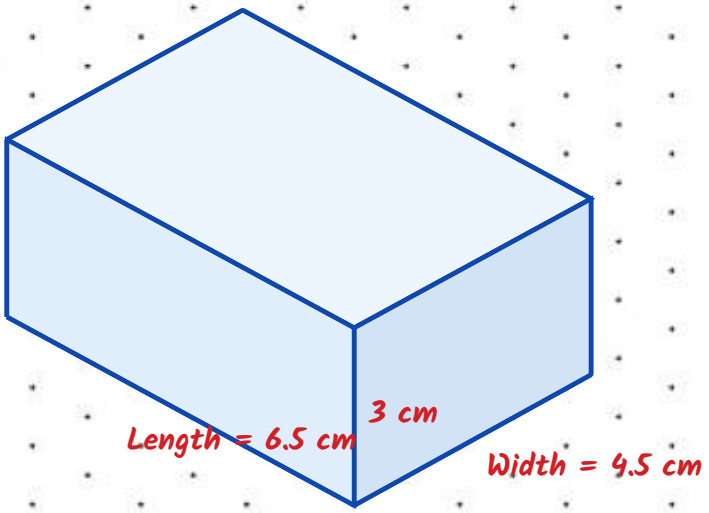
Question 18

Draw a cuboid measuring 4.5cm by 6.5 cm by 3cm. (2 marks)

- Isometric Projection: 4.5cm × 6.5cm × 3cm
- All parallel edges drawn with accurate unit counts

ber (1 cm)



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Show your calculations and/or workings out here:

Calculate 37% of 108:

Step 1: 37% = 37 ÷ 100 = 0.37

Step 2: 0.37 × 108 = 39.96 [1 mark]

Write your answer in this box:

39.96 ✓

Question 19

Calculate 37% of 108 (1 mark)

Show your calculations and/or workings out here:

Write your answer in this box:

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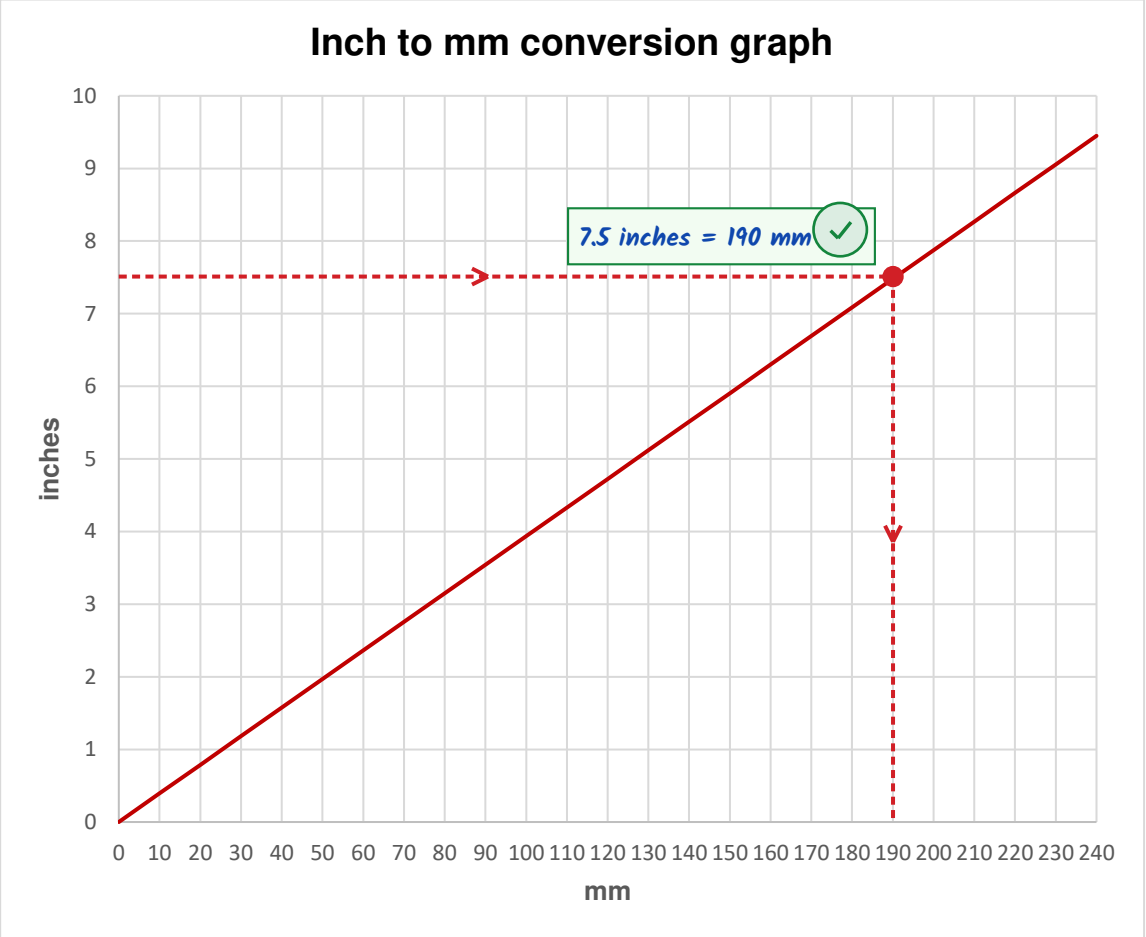
Question 20

Harry makes bracelets using round silver beads with a diameter of 4mm. He buys the beads in packets of 800.

Harry makes each bracelet to a standard bracelet length of $7\frac{1}{2}$ inches.

Using the conversion graph below, calculate how many bracelets Harry can make from one packet of beads. (5 marks)



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Show your calculations and/or workings out here:

Step 1: Read length from conversion graph for 7 1/2 inches:

7.5 inches = 190 mm (marked on conversion graph Page 31) [M1]

Step 2: Calculate number of 4mm beads per bracelet:

Beads per bracelet = 190 mm ÷ 4 mm = 47.5 beads [M1]

Requires 48 beads (or 47 beads) per bracelet [A1]

Step 3: Calculate number of bracelets from 1 packet of 800 beads:

800 ÷ 48 = 16.666... --> 16 complete bracelets [M1 A1]

(Or using 47 beads: 800 ÷ 47 = 17.02... --> 17 bracelets)

Write your answer in this box:

16 bracelets ✓

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Question 21

Harry buys his beads from a wholesale supplier. A packet of 800 beads costs £59.99.

Harry receives a discount of 18%.

How much does Harry pay **per bead** to the nearest whole pence?

(4 marks)

Show your calculations and/or workings out here:

Step 1: Calculate 18% discount on wholesale pack (£59.99):

Discount = 0.18 × £59.99 = £10.7982 [M1]

Discounted price = £59.99 − £10.7982 = £49.1918 [A1]

Step 2: Calculate cost per bead:

Cost per bead = £49.1918 ÷ 800 = £0.061489... per bead [M1]

Step 3: Round cost to nearest whole pence:

£0.061489... --> 6p (or £0.06) [A1]

Write your answer in this box:

6p (or £0.06) ✓

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Question 22

Dinah wants to send a parcel by post. She knows the parcel weighs 1lb 2oz.

Dinah finds a leaflet with information on the costs of posting parcels:

Format and max measurements	Max Weight	1st Class	2nd Class
Small parcels	100g	£1.06	83p
	250g	£1.50	£1.32
	500g	£1.97	£1.72
	750g	£2.72	£2.33
	1kg	£4.61	£4.16

Cheapest rate

How much will it cost Dinah to post the parcel using 2nd class?

(3 marks)

1kg = 2.2lbs
1oz = 28.35g

Show your calculations and/or workings out here:

Step 1: Convert parcel weight (1 lb 2 oz) into grams:

- 1 lb = 453.6 g, 2 oz = 2 × 28.35 g = 56.7 g [M1]

Step 2: Total weight = 453.6 g + 56.7 g = 510.3 g [A1]

Step 3: 510.3 g ≤ 750 g, so parcel falls into 'Up to 750g' tier.

Cheapest postage option = 2nd Class = £2.33 [A1]

Write your answer in this box:

£2.33 (2nd Class) ✓

[End of assessment]

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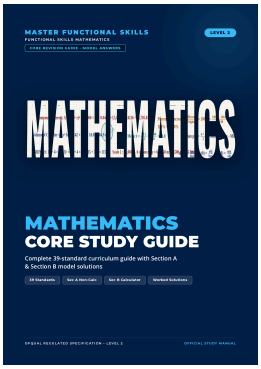

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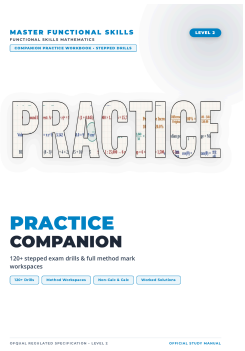

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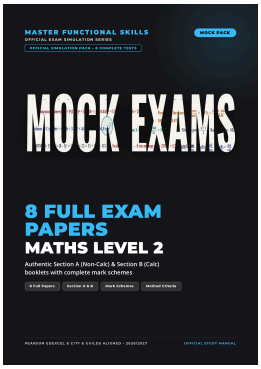

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